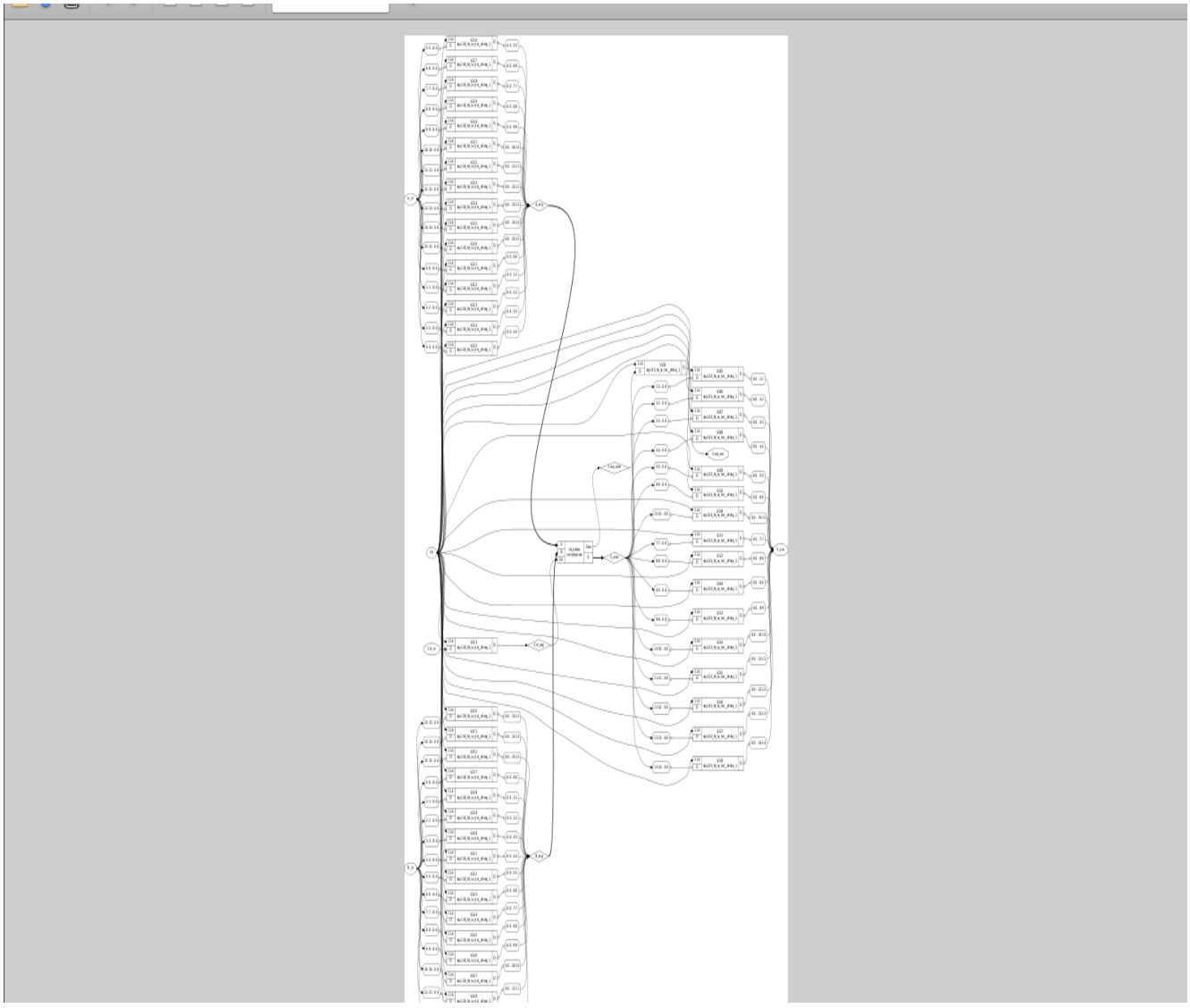
EE5311: Digital IC Design
Tutorial 11

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Ripple Carry Adder

Synthesis



Code

```
   // Top-level module with clocked registers
module carrybypass_reg(
    input clk,
    input [15:0] A_in,
    input [15:0] B_in,
    input Cin_in,
    output reg [15:0] S_out,
    output reg Cout_out
);

    reg [15:0] A_reg, B_reg;
    reg Cin_reg;

    wire [15:0] S_wire;
    wire Cout_wire;

    // Instantiate the actual carry bypass adder
    carrybypass cb_adder(
        .A(A_reg),
        .B(B_reg),
        .Cin(Cin_reg),
        .Cout(Cout_wire),
        .S(S_wire)
    );

    // Register inputs and outputs on clock edge
    always @(posedge clk) begin
        A_reg <= A_in;
        B_reg <= B_in;
        Cin_reg <= Cin_in;

        S_out <= S_wire;
        Cout_out <= Cout_wire;
    end

endmodule

// 16-bit Carry Bypass Adder composed of 4 x 4-bit adders
module carrybypass(
    input [15:0] A,
    input [15:0] B,
    input Cin,
    output Cout,
    output [15:0] S);

    wire [3:0] c;

    cbp4bit cbp0(.A(A[3:0]),     .B(B[3:0]),     .Cin(Cin),   .Cout(c[0]), .S(S[3:0]));
    cbp4bit cbp1(.A(A[7:4]),     .B(B[7:4]),     .Cin(c[0]),  .Cout(c[1]), .S(S[7:4]));
    cbp4bit cbp2(.A(A[11:8]),    .B(B[11:8]),    .Cin(c[1]),  .Cout(c[2]), .S(S[11:8]));
```

```
    cbp4bit cbp3(.A(A[15:12]),   .B(B[15:12]),   .Cin(c[2]),  .Cout(c[3]), .S(S[15:12]));

    assign Cout = c[3];
endmodule

// 4-bit Carry Bypass Block
module cbp4bit(
    input [3:0] A, B,
    input Cin,
    output Cout,
    output [3:0] S);

    wire [4:0] C;
    wire q, r;

    assign C[0] = Cin;

    fulladdr fa0(.a(A[0]), .b(B[0]), .cin(C[0]), .cout(C[1]), .Sum(S[0]));
    fulladdr fa1(.a(A[1]), .b(B[1]), .cin(C[1]), .cout(C[2]), .Sum(S[1]));
    fulladdr fa2(.a(A[2]), .b(B[2]), .cin(C[2]), .cout(C[3]), .Sum(S[2]));
    fulladdr fa3(.a(A[3]), .b(B[3]), .cin(C[3]), .cout(C[4]), .Sum(S[3]));

    bypass by0(.A(A), .B(B), .p(q));
    mux21 mu0(.i0(C[4]), .i1(C[0]), .sel(q), .o(r));

    assign Cout = r;
endmodule

// 2:1 Multiplexer
module mux21(
    input i0,
    input i1,
    input sel,
    output o);
    assign o = sel ? i1 : i0;
endmodule

// Bypass condition generator: produces propagate signal
module bypass(
    input [3:0] A,
    input [3:0] B,
    output p);
    wire [3:0] P;
    assign P = A ^ B;
    assign p = P[0] & P[1] & P[2] & P[3];
endmodule

// 1-bit Full Adder
module fulladdr(
    input a,
    input b,
```

```
    input cin,
    output cout,
    output Sum);
    assign cout = (a & b) | (a & cin) | (b & cin);
    assign Sum = a ^ b ^ cin;
endmodule
```

STA

```
 Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ _17_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.30    0.30 v _17_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.37    0.67 v cb_adder/cbp0/fa0/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.38    1.06 v cb_adder/cbp0/fa1/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.38    1.44 v cb_adder/cbp0/fa2/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.34    1.78 v cb_adder/cbp0/fa3/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.32    2.10 v cb_adder/cbp0/mu0/_4_/X (sky130_fd_sc_hd__mux2_1)
   0.38    2.48 v cb_adder/cbp1/fa0/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.38    2.87 v cb_adder/cbp1/fa1/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.38    3.25 v cb_adder/cbp1/fa2/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.34    3.60 v cb_adder/cbp1/fa3/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.32    3.91 v cb_adder/cbp1/mu0/_4_/X (sky130_fd_sc_hd__mux2_1)
   0.38    4.30 v cb_adder/cbp2/fa0/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.38    4.68 v cb_adder/cbp2/fa1/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.38    5.06 v cb_adder/cbp2/fa2/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.34    5.41 v cb_adder/cbp2/fa3/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.32    5.73 v cb_adder/cbp2/mu0/_4_/X (sky130_fd_sc_hd__mux2_1)
   0.38    6.11 v cb_adder/cbp3/fa0/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.38    6.49 v cb_adder/cbp3/fa1/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.38    6.88 v cb_adder/cbp3/fa2/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.34    7.22 v cb_adder/cbp3/fa3/_10_/X (sky130_fd_sc_hd__maj3_1)
   0.27    7.49 v cb_adder/cbp3/mu0/_4_/X (sky130_fd_sc_hd__mux2_1)
   0.00    7.49 v _16_/D (sky130_fd_sc_hd__dfxtp_1)
           7.49   data arrival time

   8.00    8.00   clock clk (rise edge)
   0.00    8.00   clock network delay (ideal)
   0.00    8.00   clock reconvergence pessimism
           8.00 ^ _16_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.12    7.88   library setup time
           7.88   data required time
---------------------------------------------------------
           7.88   data required time
          -7.49   data arrival time
---------------------------------------------------------
           0.38   slack (MET)
```

Critical Path

Critical Path Analysis

Path Sequence:

`fa0` → `fa1` → `fa2` → `fa3` → `mux2` → `fa0` → `fa1` → `fa2` → `fa3` → `mux2` → `fa0` → `fa1` → `fa2` → `fa3` → `mux2`

Data Arrival Time: 7.49 ns
Data Required Time: 7.88 ns

Slack (Setup): 0.38 ns (MET)

Clock Period: 8.00 ns
Path Type: Max (Setup Path)
Critical Path Delay: 7.49 ns